

MATH 221 Examination

Linear Algebra and Matrix Theory

Instructions:

- Answer all questions.
- For Questions 1–4, choose the best option.
- For Questions 5–8, write brief but precise answers.

1. Which property distinguishes an orthogonal matrix from a general invertible matrix?
 - (A) Its determinant equals zero
 - (B) Its transpose equals its inverse
 - (C) It has only positive eigenvalues
 - (D) Its columns are linearly dependent
2. For a system of linear equations $Ax = b$ where A is an $m \times n$ matrix with $m > n$, which condition guarantees that the least squares solution is unique?
 - (A) The matrix A has full row rank
 - (B) The matrix A has full column rank
 - (C) The vector b is in the column space of A
 - (D) The null space of A is trivial and $m = n$
3. Given a symmetric matrix S , which of the following statements is always true?
 - (A) All eigenvalues are positive
 - (B) All eigenvalues are real
 - (C) All eigenvectors are orthogonal to each other regardless of eigenvalue multiplicity
 - (D) The matrix is always invertible
4. What is the primary geometric interpretation of the singular value decomposition (SVD) of a matrix $A = U\Sigma V^T$?
 - (A) It represents A as a rotation followed by a scaling
 - (B) It decomposes A into a sequence of rotations, scaling, and another rotation
 - (C) It expresses A as a sum of rank-one matrices
 - (D) It diagonalizes A in its own basis

5. Explain why the dimension of the null space and the dimension of the column space of a matrix A must sum to the number of columns of A . What fundamental theorem supports this?
6. Consider the Gram-Schmidt process for orthogonalizing a set of linearly independent vectors. What potential numerical issues arise when implementing this algorithm in practice, and how can modified Gram-Schmidt address these concerns?
7. Describe the relationship between the eigenvalues of a matrix A and the eigenvalues of $A^T A$. Why is this relationship important in the context of the singular value decomposition?
8. Explain why the condition number of a matrix, defined as $\kappa(A) = \|A\| \|A^{-1}\|$, provides a measure of how sensitive the solution to $Ax = b$ is to perturbations in b or A . What happens when $\kappa(A)$ is very large?